

Camouflage Cloak

```
# Import Libraries
```

```
import numpy as np
```

```
import cv2
```

```
import time
```

```
# camCount=0
```

```
# To use webcam enter 0 and to enter the video path in double quotes
```

```
cap = cv2.VideoCapture(0)
```

```
time.sleep(
```

```
    3) # paranthesis has two because the camera needs time to adjust it  
    self i according to the environment(ANDHERA KAMRA)
```

```
background = 0
```

```
# Capturing the background
```

```
for i in range(60):
```

```
    ret, background = cap.read()
```

```
# capturing image
```

```
background = np.flip(background, axis=1)
```

```
while (  
cap.isOpened()): # Condition for this is when only the web cam is  
opened it will only run the code else the code will not run in the  
background without the webbcam
```

```
    ret, img = cap.read()
```

```
    if not ret:
```

```
        break
```

```
    img = np.flip(img, axis=1)
```

```
    hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
```

```
    # HSV values
```

```
    # setting the values for the cloak
```

```
    lower_red = np.array([0, 120, 70])
```

```
    upper_red = np.array([10, 255, 255])
```

```
    mask1 = cv2.inRange(hsv, lower_red, upper_red)
```

```
    lower_red = np.array([170, 120, 70])
```

```
    upper_red = np.array([180, 255, 255])
```

```
    mask2 = cv2.inRange(hsv, lower_red, upper_red)
```

```
    mask1 = mask1 + mask2
```

```
    mask1 = cv2.morphologyEx(mask1, cv2.MORPH_OPEN,  
np.ones((3, 3), np.uint8), iterations=2)
```

```
mask1 = cv2.morphologyEx(mask1, cv2.MORPH_DILATE,  
np.ones((3, 3), np.uint8), iterations=1)
```

```
mask2 = cv2.bitwise_not(mask1)
```

```
res1 = cv2.bitwise_and(background, background, mask=mask1)
```

```
res2 = cv2.bitwise_and(img, img, mask=mask2)
```

```
final_output = cv2.addWeighted(res1, 1, res2, 1, 0)
```

```
cv2.imshow('Invisible Cloak', final_output)
```

```
k = cv2.waitKey(10)
```

```
if k == 27:
```

```
    break
```

```
cap.release()
```

```
cv2.destroyAllWindows()
```